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FEDERAL SPECIFICATION

SURFACERS (PLANERS) WOODWORKING

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric motor driven woodworking surfacers, for planing wood workpieces, to produce surfaces at a determined thickness, within the dimensional workpiece capacity of the surfacer.

1.2 Classification.

1.2.1 Types, classes and sizes. The surfacers shall be of the following types, classes, and sizes. The type, class, and size surfacer required shall be as specified (see 6.2).

Type I - Single cutterhead.

Class A - Light duty, finishing.

Size 12 by 5 - 12-inch wide by 5-inch thick workpiece capacity.
Size 18 by 6 - 18-inch wide by 6-inch thick workpiece capacity.
Size 24 by 8 - 24-inch wide by 8-inch thick workpiece capacity.

Class B - Medium duty, finishing.

Size 24 by 8 - 24-inch wide by 8-inch thick workpiece capacity.
Size 30 by 8 - 30-inch wide by 8-inch thick workpiece capacity.
Size 36 by 8 - 36-inch wide by 8-inch thick workpiece capacity.

Class C - Heavy duty, finishing.

Size 30 by 8 - 30-inch wide by 8-inch thick workpiece capacity.
Size 36 by 8 - 36-inch wide by 8-inch thick workpiece capacity

FSC 3220

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Type II - Double cutterheads.

Class C - Heavy duty, finishing.

Size 30 by 7 - 30-inch wide by 7-inch thick workpiece capacity.

Class D - Heavy duty, roughing.

Size 30 by 14 - 30-inch wide by 14-inch thick workpiece capacity.

Class E - Heavy duty, combination, finishing and roughing.

Size 18 by 8 - 18-inch wide by 8-inch thick workpiece capacity.
Size 24 by 8 - 24-inch wide by 8-inch thick workpiece capacity.
Size 30 by 8 - 30-inch wide by 8-inch thick workpiece capacity.
Size 36 by 8 - 36-inch wide by 8-inch thick workpiece capacity.
Size 42 by 8 - 42-inch wide by 8-inch thick workpiece capacity.
Size 36 by 12 - 36-inch wide by 12-inch thick workpiece capacity.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Regulation:

Occupational Safety and Health (OSHA) 29 CFR 1910

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on subscription basic by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When reinstated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Copies of OSHA standard and the standards which are incorporated by reference may be examined at the National Office of the Occupational Safety and Health Administration, U.S. Department of Labor, Washington, DC 20210.

Federal Standards:

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal

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Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Military Standards:

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717).

American National Standards Institute (ANSI):

ANSI 01.1 - Safety Requirements for Woodworking Machinery.

(Application for copies should be addressed to the American National Standard Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

National Electrical Manufacturers' Association (NEMA):

ANSI/NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

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American Society For Testing and Materials (ASTM):

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

National Fire Protection Association (NFPA):

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.2 Design. The surfacer shall be new (not a prototype) and one of the manufacturer's current product models capable of operations in accordance with the requirements herein. The surfacer shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Design and construction dimensional systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the surfacer. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than .01 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

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3.2.3 Energy efficiency. The surfacer and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the surfacer, whether or not specifically required herein, shall comply with all requirements of 29 CFR 1910 and ANSI 01.1. If a conflict arises between 29 CFR 1910 and ANSI 01.1, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified in detail (see 6.2.1).

3.2.5.1 Safety guarding. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial surfacers supplied to the commercial market shall be provided for the point of operation and all other parts of the surfacer that present safety hazards. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the parts. The safety devices shall not interfere with the operation of the surfacers. If specific point-of-operation guarding dependent on workpiece size and configuration is required, the procuring activity shall specify the exact configuration of the guard required (see 6.2.1).

3.2.6 Mercury restriction. The surfacer shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the surfacer.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the surfacer shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. Means shall be provided to indicate a low-oil condition. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3. Construction. The surfacer shall be constructed of parts which are new, without defects, and free or repairs. The structure shall withstand all forces encountered during operation of the surfacer to its maximum rating and capacity without permanent distortion.

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3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the surfacer shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts, either US or SI, used on the surfacer and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Surfacers covered herein shall consist essentially of a frame, tables, feed rolls, cutterhead, cutterhead drive, platen, chipbreaker, pressure bars, and such other components necessary to meet the requirements stated herein.

3.4.1 Frame. The frame shall be of such mass and shall have sufficient strength and rigidity to support the working parts of the surfacer, while surfacing maximum width workpieces to a desired thickness. The frame shall support the cutterhead assembly, including its alignment to the tables and platen, to meet the performance requirements specified herein. The frame shall be constructed with ways, of either the inclined wedge bed type, or vertical screw bed type, with ways on the tables providing adjustment for varying thickness of workpieces within the capacity of the surfacer. The frame shall have not less than three holes for mounting to the floor or foundation. The frame shall have features as necessary to provide for connection to a dust and chip collection system.

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3.4.2 Tables. The surfacer shall have an infeed or front table and an outfeed or rear table with two or more table rolls. Type I surfacers shall have a center table or platen. The tables, platen, and rolls shall have vertical adjustments to encompass the entire size workpiece range of the surfacer. The ways supporting the table shall be fitted with adjustable gibs for maintaining table alignment and working tolerances. Type I, class A surfacers shall have a scale, graduated in increments of 1/8 inch or less, with each inch increments having a longer scribe mark and bearing a number. The scale shall be permanently attached to the surfacer. Type I, classes B and C, and type II surfacer shall have a micrometer adjusting device with a corresponding scale to facilitate table adjustments. Unless otherwise specified (see 6.2.1), type I, class C, and type II surfacers shall have either a lever or push buttons for powered control of table adjustments. The powered table positioning device shall have means for preventing the tables and platen from contact with the cutterhead knives.

3.4.3 Cutterhead. The cutterhead assembly shall be supported by ball or roller bearings of a size, type, and class suitable for the application. The cutterhead shall have knives extending across the entire cutterhead length. The number of knives in the cutterhead shall be not less than the number of knives required, by surfacer size, in table I. Unless otherwise specified (see 6.2.1), the knives shall be manufactured from tool steel. When specified (see 6.2.1), the knives shall be manufactured from carbide steel in lieu of tool steel. The cutterhead assembly shall have means for precisely locating, and securely holding the knives, in uniformly adjusted positions. When specified (see 6.2.1 and 6.4), a special cutterhead assembly shall be mounted on the surfacer. Special cutterheads shall be constructed to accommodate either spiral type carbide knives extending, around and across, the entire cutterhead length, or carbide knives shorter than cutterhead length, which are located in overlapping positions, in a helical pattern around the cutterhead, assuring a uniform cut the entire cutterhead length. Special cutterheads having knives shorter than cutterhead length, also consist of indexable and replaceable carbide knives, not wider than 1 inch, having not less than four sides. Each side of the carbide knife shall have a cutting edge. Each indexable knife shall be supported in a knife holder, which is inserted into the cutterhead and, providing adjustment of knife height to obtain a uniform cut over the entire cutterhead length. Unless otherwise specified (see 6.2.1), cutterhead speeds shall be not less than the direct drive and belted drive speeds of table I.

3.4.3.1 Cutterhead drive. Unless otherwise specified (see 6.2.1), the cutterhead drive for type I, class A surfacers shall either be direct or belt driven. Unless otherwise specified (see 6.2.1), the cutterhead drive for type I, classes B and C, and type II, classes C and D surfacers shall be direct driven. Unless otherwise specified (see 6.2.1), type II, class E surfacers shall be belt driven. Belt driven surfacers shall have means of maintaining belt tension and alignment. Belt drives shall be capable of transmitting full motor torque without slippage.

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3.4.4 Rolls. Each surfacer shall have vertically adjustable, smooth table rolls between the infeed and the outfeed tables. The vertical adjustment of the table rolls shall be not less than 0.005 inch above the table surface. When specified (see 6.2.1), the infeed table roll and the outfeed table roll height shall be controlled by either a single manually operated lever or a single micrometer type adjustment device, graduated in increments not greater than 0.005 inch over an adjustment range of not less than 0.030 inch. Each surfacer shall have corrugated infeed and smooth outfeed rolls, mounted in a suspension device in the upper frame housing. Rolls in the upper frame shall have means of vertical adjustment, for setting roll pressure, including vertical movements required for varying thickness in a workpiece. Type I, class A, sizes 12 by 5 and 18 by 6 shall have a solid-corrugated upper infeed roll. All other surfacers shall have corrugated, upper infeed rolls of the sectional type, having individual section yield, with each section having a dimensional width of 3 inches or less.

3.4.5 Feed drive. Unless otherwise specified (see 6.2.1), the surfacer shall have an infinitely variable feed range conforming to table I. Unless otherwise specified (see 6.2.1), feed rolls on type I, classes B and C, and type II surfacers shall be independently powered by a separate motor; type I, class A feed rolls shall either be independently driven or power driven from the cutterhead motor. Unless otherwise specified (see 6.2.1), type I, classes A and B, and type II, class D surfacers shall have not less than one upper infeed and one upper outfeed roll, and two table rolls; type I, class C surfacers shall have not less than two upper infeed rolls; type I, class C surfacers shall have not less than two upper infeed rolls, one outfeed roll and three table rolls; type II, class C surfacers shall have not less than two upper infeed rolls, two upper outfeed rolls and four table feed rolls. In addition to feed rolls and table rolls, sectional and solid, type II, class E surfacers shall have an independent driven, endless feed chain assembly suspended in the upper frame housing, for guiding and feeding workpieces over the tables between the cutterheads. The feed chain assembly shall be constructed to allow for guiding and feeding workpieces having warp or twisted conditions which would normally split if surfaced through solid upper and lower rolls. The feed chain assembly shall consist essentially of shafts, sprockets and links. The links shall support a vertical yielding type device, with hardened steel edges or comparable material, that contacts the surface of the workpiece. The device contacting the surface of the workpiece shall be so shaped that a restraining force to side movement is imposed while feeding the workpiece through the surfacer. The vertical movement of the yielding device shall have a yield range, and be positioned on the links, to positively feed workpieces not longer than 18 inches, upon feeding them singularly, through the surfacer.

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3.4.6 Chipbreakers. Each surfacers shall have a chipbreaker preceding the cutterhead to effectively direct the flow of chips and maintain pressure of workpieces being surfaced. Type I, class A surfacers shall have chipbreakers of either the individually yielding sectional type or of the solid type. Chipbreakers on type I, classes B and C and upper cutterhead type II, class C surfacers shall be of the individually yielding, sectional type, not more than 2 inches in width, and shall have a yield range of not less than 1/4 inch. Type II, class D upper cutterhead chipbreakers shall have sectional width of not less than 3 inches with a yield range of not less than 1/4 inch. Type II, class E upper cutterhead chipbreakers shall have sectional width of not less than 2 inches with a yield range of not less than 3/16 inch.

3.4.7 Pressure bars. Each cutterhead, except lower cutterhead type II, class D, shall have a pressure bar directly behind and with the bottom or lower portion concentrically shaped with the cutting circle. Pressure bars shall be constructed and mounted to provide vertical adjustment and maintain pressure on workpieces being surfaced. The pressure bar for type I, class C and the pressure bar on the upper cutterhead for type II, class C surfacers, shall be either plated with hard chromium or other materials having equal resistance to corrosion and wear.

3.4.8 Grinding and jointing attachment. Unless otherwise specified (see 6.2.1), a knife blade grinding and jointing attachment shall be furnished with each cutterhead. The attachment shall consist essentially of a head constructed to accommodate both an electric motor driven grinding wheel and a jointing stone, and a mechanism that allows the grinding stone and the jointing stone to be manually traversed across the entire length of the cutterhead. The attachment shall also have means for precisely adjusting the wheel and stone in relationship to the knives, and a means for retaining adjustment during the grinding and jointing operations. When carbide knife blades are specified (see 3.4.3), the grinding wheel shall be suitable for grinding carbide steel knife blades, and the jointing stone described above is not required to be furnished.

3.4.9 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Unless otherwise specified (see 6.2), the surfaces shall initially wired for operation on 460 volts.

3.4.9.1 Motors. Motors shall have ball or roller bearings. Unless otherwise specified (see 6.2.1), continuous duty horsepower ratings shall conform to the requirements of table I. When specified (see 6.2.1), the surfacer shall be equipped with a motor having the optional horsepower stated in table I. Unless otherwise specified (see 6.2.1), each cutterhead motor shall be single speed and meet the requirements of a total enclosure, and each feed motor shall meet the requirements of a drip-proof enclosure. All meters shall conform to the requirements of NEMA MG-1.

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3.4.10 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each surfacer:

(a) One set of hand tools normally furnished by the manufacturer.

(b) One extra set of cylinder knives.

3.4.11 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the surfacer shall be not less than specified in table I for this surfacer size specified.

3.6. Performance requirements.

3.6.1 Material removal rate and accuracy. The surfacer shall be capable of surfacing workpieces of soft and hardwood from 1/8-inch thickness to include the capacity, workpiece width and thickness, requirements of table I. The surfacer, when connected to an appropriate dust and chip collection system, shall be capable of removing not less than 3/16 inch from maximum width, wood workpieces, producing a surfaced workpiece free of chip marks, glazed areas, chatter marks and sniping. Finished surfaces are not to be expected from roughing operations.

3.7 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.7.1 Lubrication plate. A lubrication plate shall be permanently and securely attached to each machine. Unless otherwise specified (see 6.2.1), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Military of Federal specification number for each lubricant
Size and type of each lubricant filter

3.7.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.8 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.9 Workmanship. Workmanship of the surfacer shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.7).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to acceptance of defective material.

4.1.2 Classification of inspections. When specified, the inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Quality conformance inspection (see 4.2.1).

c. Acceptance test (see 4.2.2).

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4.1.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, test and examinations shall be performed in an indoor facility having a temperature of from 50 to 100 degrees Fahrenheit (F), and up to 95 percent humidity.

4.2 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspections shall consist of the examination in 4.3 and all tests in 4.4. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.2.1 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspections shall consist of the examination in 4.3, the tests in 4.4, and the inspection in 4.5. Failure of the item to pass the examination, the test, or the inspection shall be cause for rejection.

4.2.2 Acceptance test. Acceptance test shall be performed on the surfacer to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the surfacer is installed at its final location. The acceptance test shall consist of the examination in 4.3 and all tests in 4.4. Failure of the surfacer to pass one or more tests or any part of a test will be cause for rejection.

4.3 Examination. The surfacer shall be examined to determine compliance with all requirements of this specification including compliance with all applicable standards cited in section 2.

4.4 Tests.

4.4.1 Operational test. The surfacer shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.4.2 Performance test. The surfacer shall be subjected to material removal rate and accuracy in 3.6.1, and the following test. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1). All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.4.2.1 Test workpiece. The type of hardwood used as a test workpiece shall be at the option of the supplier. The test workpiece dimension shall be not less than 1 inch in thickness, 6 feet in length and either of a width equal to the length of the cutterhead or, two test workpieces, with a workpiece at each end of the cutterhead. Type I surfacers shall have the workpiece turned over after surfacing one side; then the opposite side shall be surfaced. Type II surfacers shall remove material from the top and bottom surfaces in one pass.

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4.4.2.2 Workpiece examination. The finished surfaces of the test workpiece shall be free of chip marks, glazed areas, chatter marks and sniping. The test workpiece thickness shall be uniform within ±0.005 inch.

4.4.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

4.5 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packaging, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The surfacers covered by this specification are intended for surfacing and sizing, in thickness, of wood workpieces, in woodworking shapes.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If surfacer is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Additional safety and health requirements, if required (see 3.2.5).

f. Painting, if different (see 3.3.5).

g. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

h. Type table adjustment if different (see 3.4.2).

i. Carbide knife blades, if required (see 3.4.3).

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j. Specify either spiral type knives, knives shorter than cutterhead length, or indexable helically pattern cutterhead and knives, if required (see 3.4.3 & 6.4).

k. Cutterhead speed if different (see 3.4.3).

l. Type cutterhead drive if different (see 3.4.3.1).

m. Single micrometers type table roll adjustment, if required (see 3.4.4).

n. Feed range, if different (see 3.4.5)

o. Feed roll drive if different (see 3.4.5).

q. Configuration of feed rolls if different (see 3.4.5)

r. Knife grinding and jointing attachment if not required (see 3.4.8).

s. Electrical system and input power, if different (see 3.4.9).

t. Initial voltage, if different (see 3.4.9).

u. Horsepower rating if different (see 3.4.9.1).

v. Optional horsepower when required (see 3.4.9.1).

w. Motor enclosure, if different (see 3.4.9.1).

x. Standard equipment if different (see 3.4.10).

y. Specify and fully describe optional equipment, if required (see 3.4.11).

z. Size and capacity, if different (see 3.5).

aa. Lubrication chart or plate, if different (see 3.7.1).

bb. Nameplate, if different (see 3.7.2).

cc. Inspection conditions, if different (see 4.1.3).

dd. First article inspection, if different (see 4.2).

ee. Quality conformance inspection, if different (see 4.2.1).

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ff. Acceptance test, if different (see 4.2.2).

gg. Performance test, if different (see 4.4.2).

hh. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should be include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Special knife blades and cutterhead assemblies. Due to the relatively high cost of carbide, spiral, and helical knives and cutterheads (see 3.4.1.3, 6.2.1(h), and 6.2.1(i), procuring agencies are cautioned to carefully evaluate their requirements for these special cutters. The items should be required only for heavy duty use where the higher initial cost can be amortized by extended cutter life. In addition, requiring this special equipment might eliminate the commercial equipment of some potential suppliers.

6.5 Supersession data.

6.6 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Safety and health requirements. Standard 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity is advised to analyze the existing harzard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Inspection facilities. The contracting officer should be notified by each contractor in his response to the solicitation of the location where inspection tests are to be performed. It is recommended all inspections be performed within the limits of the continental United States.

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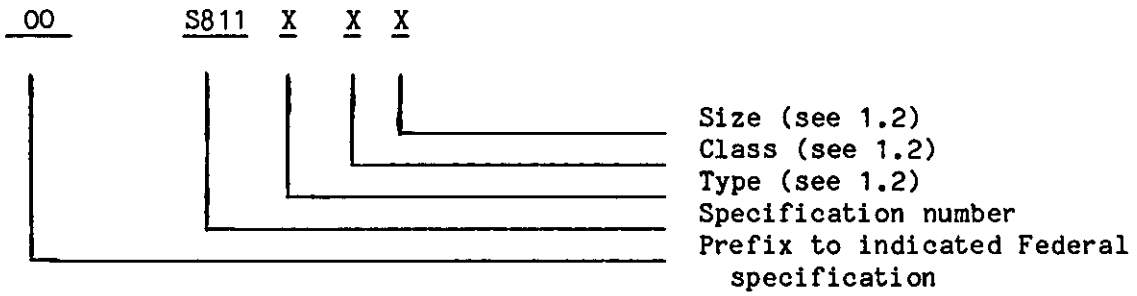
6.9 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.10 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.11 Subject term keyword) listing.

Double cutterhead
Single cutterhead
Surfacers
Woodworking

6.12 Part or identifying number(s) (PINs). The PINs to be used for items acquired to this specification are created as follows:



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MILITARY INTERESTS:

Custodians

Army - ME
Navy - SH
Air Force - 99

Review Activities

Army - ME
Navy - YD, MC
DGSC - GS
Air Force - 84

User Activities

Army - CE
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

DOD Project 3220-0153

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
00-S-811D	.Surfacers (Planers) Woodworking

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: –

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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